

| Assessment | Test | Domain                            | Skill                                                | Difficulty                                          |
|------------|------|-----------------------------------|------------------------------------------------------|-----------------------------------------------------|
| SAT        | Math | Problem-Solving and Data Analysis | Inference from sample statistics and margin of error | Medium <div><div></div><div></div><div></div></div> |

Question

A bag containing 10,000 beads of assorted colors is purchased from a craft store. To estimate the percent of red beads in the bag, a sample of beads is selected at random. The percent of red beads in the bag was estimated to be 15%, with an associated margin of error of 2%. If  $r$  is the actual number of red beads in the bag, which of the following is most plausible?

Answer

- A.  $r > 1,700$
- B.  $1,300 < r < 1,700$
- C.  $200 < r < 1,500$
- D.  $r < 1,300$

Correct Answer: B

Rationale

Choice B is correct. It was estimated that 15% of the beads in the bag are red. Since the bag contains 10,000 beads, it follows that there are an estimated  $10,000 \times 0.15 = 1,500$  red beads. It's given that the margin of error is 2%, or  $10,000 \times 0.02 = 200$  beads. If the estimate is too high, there could plausibly be  $1,500 - 200 = 1,300$  red beads. If the estimate is too low, there could plausibly be  $1,500 + 200 = 1,700$  red beads. Therefore, the most plausible statement of the actual number of red beads in the bag is  $1,300 < r < 1,700$ .

Choices A and D are incorrect and may result from misinterpreting the margin of error. It's unlikely that more than 1,700 beads or fewer than 1,300 beads in the bag are red. Choice C is incorrect because 200 is the margin of error for the number of red beads, not the lower bound of the range of red beads.